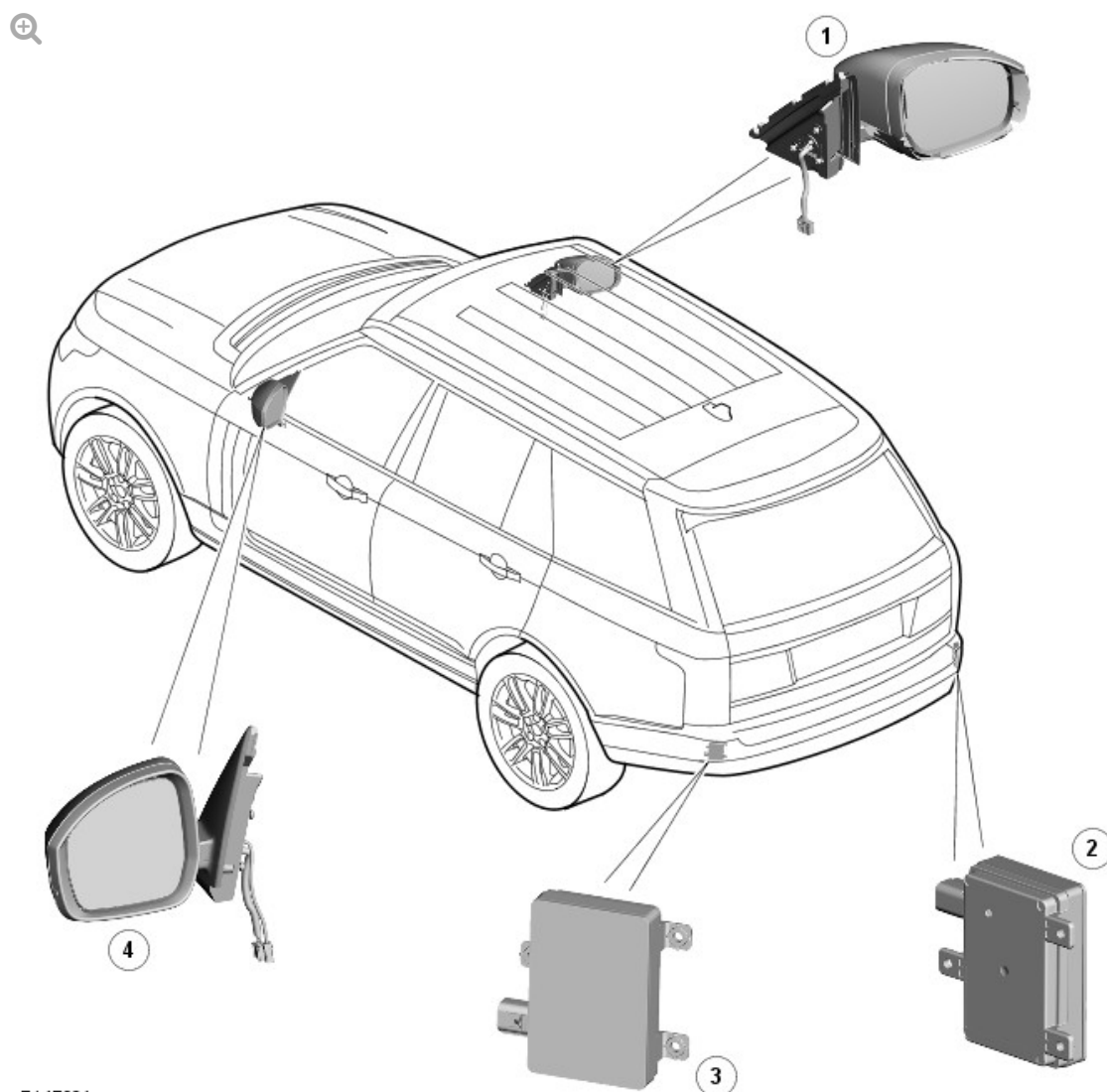


2016.0 RANGE ROVER (LG), 413-09

WARNING DEVICES

DESCRIPTION AND OPERATION

COMPONENT LOCATION



E147964

ITEM

DESCRIPTION

1	Right Side Door Mirror Warning Lamp
2	Right Side Blindspot Monitoring Control Module (BMCM)
3	Left Side Blindspot Monitoring Control Module (BMCM)
4	Left Side Door Mirror Warning Lamp

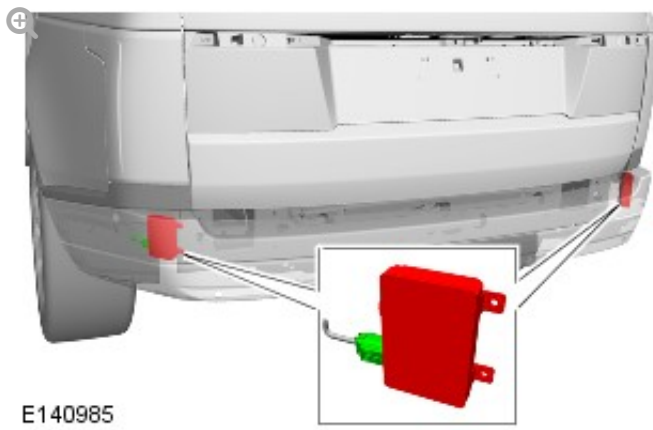
OVERVIEW

Eliminating blind spots is a major element in vehicle body design, but because of the structural requirements of B, C and D pillars, blind spots cannot be entirely eliminated, so the blind spot monitoring is there to help assist the driver safely crossover in moving traffic.

Blindspot monitoring comprises:

- Left side Blindspot Monitoring Control Module (BMCM)
- Right Side Blindspot Monitoring Control Module (BMCM)
- Left Side Door Mirror Assembly
- Right Side Door Mirror Assembly

The system uses two radar sensors operating at a frequency of 24GHz, they are incorporated in each of a BMCM (blindspot monitoring control module), that are located at the rear of the vehicle (one module each side) behind each of the rear wheels and are connected via a private HS (high speed) CAN (controller area network) bus to exchange information.



These modules will warn the driver of the presence of vehicles in a defined closing vehicle warning zone, or if a vehicle enters the blind spot warning zone, which poses a threat to the driver if a lane change manoeuvre is made.

Each BMCM (blindspot monitoring control module) has an 8 way connector that has power, ground, MS (medium speed) CAN comfort bus wires, two private CAN wires and two hard wire connections one to the status light and one for the alert icon in the mirror.

The BMCM carries out an auto-alignment self-check every time when in use, this procedure can take up to an hour in some environments, however the operation of the blindspot monitoring system will continue to operate as normal.

Each module will detect a vehicle in the driver's blindspot. Once the vehicle is detected the module illuminates an amber warning 'alert icon' LED (light emitting diode) in the relevant exterior door mirror.

If the system displays 'blind spot not available' and there is a misalignment trouble code in the system, the physical sensors must be checked for mounting and body work should be checked for damage or repair.

When the system initiates, it performs a self-check, during which the warning icons in the mirrors illuminate alternately for a short period of time. Each module does a left/right determination check when the ignition is

switched on. Each mirror has a different circuit configuration so that the modules can determine which mirror they are connected to. If a module detects the wrong mirror it will go into a fault condition.

The BMCM receives vehicle speed on the MS CAN comfort bus from the ABS (anti-lock brake system) control module via the GWM (gateway module) and is active from 8mph (13kph) down to 3mph (5kph). The system returns to standby state once the speed drops below 3mph (5kph).

CAUTION:

The blind spot monitoring system is designed as a driver aid not a safety device. The driver should always exercise due care and attention whilst driving.

OPERATION

The blindspot monitoring uses a cross over radar pattern to the rear of the vehicle and is used to determine actual moving vehicles eliminating any stationary roadside objects which are false warnings.

The two radar sensors mounted at the rear of the vehicle will continuously scan for and track traffic on the areas to the left-rear and right-rear of the vehicle and they can also track vehicles from as far as 73 meters.

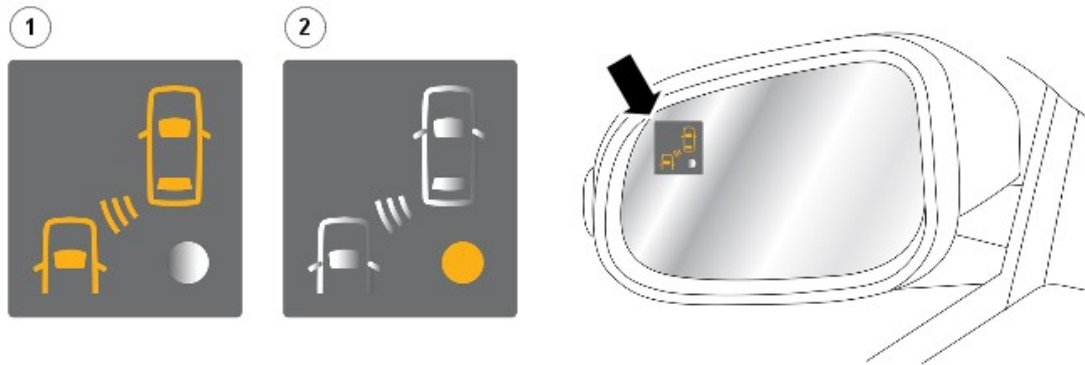
They will warn the driver of the presence of vehicles in a defined closing vehicle warning zone, which poses a threat to the driver if a lane change manoeuvre is made.

The driver will be notified of the threat by flashing the existing blind spot icon in the relevant door mirror.

NOTE:

If an overtaking vehicle is detected on both sides of the vehicle simultaneously, the warning alert icons in both mirrors will illuminate.

The blindspot monitoring lenses are shaped so as to minimize the visibility to other drivers. The LED's are located towards the outside extremity of the mirror face, within the peripheral view of the driver but not in any area of the mirror where they could obscure or distract from the image.



E97753

ITEM	DESCRIPTION
1	Warning alert icon
2	System status warning indicator

The LED lighting sequence is as follows;

- Amber alert LED icon permanently lit - system operational, vehicle detected in blind spot area.
- No LED lit – system active no vehicle detected in blind spot area.
- Amber status LED permanently lit - system not active or faulty.

The system has operating limitations and is automatically turned off under

certain operating conditions. During these operating conditions the amber status LED is permanently lit.

The system operating limitations are as follows;

- The system is inactive until vehicle speed is greater than 8mph – 13kph (amber status LED permanently lit).
- The system is inactive if an approved trailer is connected to the vehicle (amber status LED permanently lit).
- The system is inactive when reverse gear or park is selected (amber status LED permanently lit).

If either of the radar signals are blocked or distorted, for example by water, the radar face of the module is covered in mud, sleet or snow the system may detect this and be disabled with the amber status LED permanently lit together with a 'blind spot monitoring blocked' message displayed in the instrument cluster message center. The system is disabled until the blockage is cleared.

If there is a fault in the system the amber status LED is permanently lit and a 'blindspot monitoring not available' message displayed in the IC (instrument cluster) message center. The system is disabled until the fault is rectified.

System fault and blockage warnings are as follows;

- The system is disabled when the radar module signal is blocked (amber status LED permanently lit and instrument cluster message).
- The system is disabled by a fault (amber status LED permanently lit and instrument cluster message).

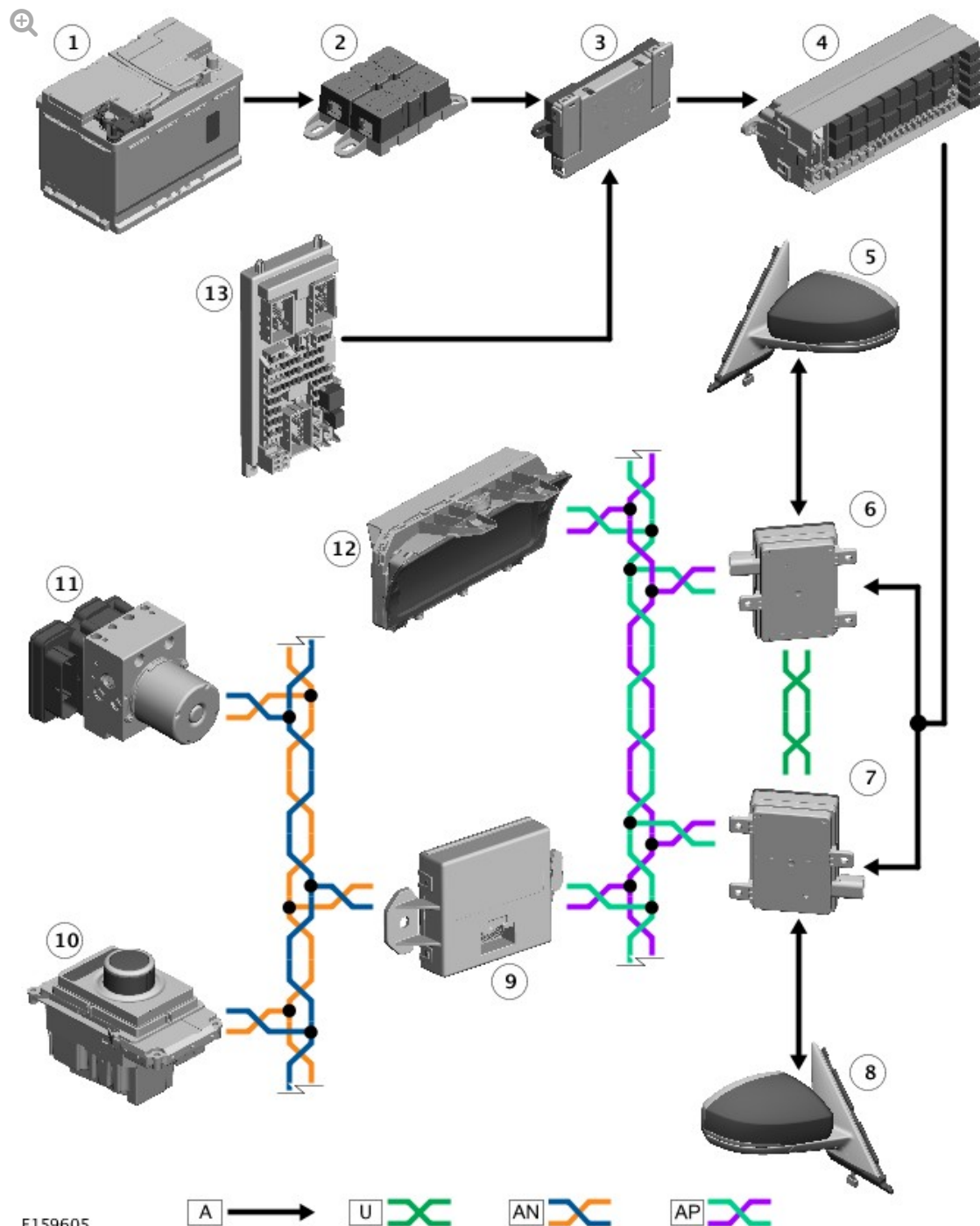
If there is a failure in the communication network and the warning LED cannot be displayed in the mirror, a failure message will be displayed in the instrument cluster message center.

When any faults are present in the system, DTC (diagnostic trouble code)'s are stored in both BCM. Replacement of BCM requires configuring

using the Land Rover approved diagnostic equipment.

Calibration of the modules using the Land Rover approved diagnostic equipment enables updates to be downloaded as new technology becomes available or any fault concerns require software updates.

CONTROL DIAGRAM



E159605

A = HARDWIRED; U = PRIVATE CAN BUS; AN = HS (HIGH SPEED) CAN
POWERTRAIN BUS; AP = MS (MEDIUM SPEED) CAN COMFORT BUS.

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box 2 (BJB2)
3	Battery Junction Box (BJB)
4	Rear Junction Box (RJB)
5	Left Side Door Mirror
6	Left Side Blindspot Monitoring Control Module (BMCM)
7	Right Side Blindspot Monitoring Control Module (BMCM)
8	Right Side Door Mirror
9	Gateway Module (GWM)
10	Transmission Control Switch (TCS)
11	Anti-lock Braking System (ABS) control module
12	Instrument Cluster (IC)
13	Central Junction Box (CJB)